Autonomous Disproof of the Erdős Unit Distance Conjecture by a GPT-5.5 Pro Agent*

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Abstract

OpenAI's recent disproof of the Erdős unit distance conjecture represents a milestone in using artificial intelligence to solve important open mathematical problems. However, this achievement relied on an internal model with undisclosed computational costs, limiting independent reproduction and analysis by the broader research community. In this paper, we present a simple autonomous agent built upon the publicly available GPT-5.5 Pro. Using a problem-agnostic, three-stage prompting pipeline—brainstorm, execute, and review—the agent consistently disproved the conjecture, generating a correct proof in 8 out of 8 independent trials. These 8 proofs differ from existing AI-generated proofs. The system operates efficiently, consuming an average of approximately 107k reasoning tokens per trial. We open-source our code and outputs, providing the first publicly accessible AI agent to autonomously resolve this conjecture.

1 Introduction

In May 2026, OpenAI announced [1] that an internal AI model autonomously disproved the Erdős unit distance conjecture without human intervention.[1] This achievement represents a milestone in the application of artificial intelligence to mathematics, demonstrating that AI systems are now capable of resolving decades-old open research conjectures.

The Erdős Unit Distance Conjecture. The mathematical problem at the center of this breakthrough was introduced by Paul Erdős in 1946 [3]. It asks for the maximum number of unit-distance pairs, $u(n)$, that can occur within a set of n points in the Euclidean plane. Erdős demonstrated a lower bound of $n^{1+\Omega(1/\log\log n)}$ using a scaled square grid, and conjectured that this bound was asymptotically tight, predicting an upper bound of $n^{1+o(1)}$. For decades, the mathematical community largely believed Erdős's conjecture to be

*Project repository: `https://github.com/yichenhuang/unit-distance`

[1] As noted in [2], Lijie Chen operated the model; Mark Sellke and Mehtaab Sawhney verified the proof.

true. In his original paper, Erdős also established an elementary upper bound of $O(n^{3/2})$ by observing that the unit distance graph cannot contain the complete bipartite graph $K_{2,3}$ as a subgraph, since two unit circles intersect in at most two points [3]. The best known upper bound, $O(n^{4/3})$, was proved by Spencer, Szemerédi, and Trotter in 1984 [4], with a subsequent shorter proof by Székely in 1997 using crossing numbers [5].

The recent AI-generated proof [6] revealed that the conjecture is false, demonstrating the existence of an absolute constant $\delta > 0$ such that $u(n) \geq n^{1+\delta}$ for infinitely many positive integers n. Rather than using traditional geometric or combinatorial methods directly in two dimensions, the proof relies on algebraic number theory. It works by constructing highly structured lattices in higher-dimensional spaces using infinite sequences of number fields. By projecting these high-dimensional structures down to the two-dimensional plane, the proof yields point sets with the required abundance of unit distances. Subsequent human analysis simplified the AI's argument [7], and a further refinement explicitly bounded the constant, yielding the lower bound of $u(n) \geq n^{1.014}$ for infinitely many n [2].

Subsequent Developments. Following OpenAI's breakthrough, Levent Alpoge [8] announced that an agent based on Anthropic's Claude Mythos autonomously solved the conjecture, yielding a different proof. However, because neither of these models is publicly available, the broader research community lacks the access needed to independently reproduce these autonomous runs. Furthermore, while OpenAI published data regarding its model's robustness (Figure 1), the absolute amount of test-time compute required remains undisclosed. For the Claude Mythos agent, both the robustness and computational cost are entirely unknown.

Attempts to solve the conjecture using publicly available models have also been made. Recently, Xiao Ma generated a proof using GPT-5.5 Pro [9]. However, the published transcript [10] documents an interactive process where the human user evaluates the model's intermediate outputs and selects specific mathematical directions to pursue. Although Ma argued that this manual guidance could in principle be automated given sufficient test-time compute, the published transcript in its present form represents a human-in-the-loop approach. Consequently, a fully autonomous solution using a public model has not yet been demonstrated.

Our Contribution. In this paper, we present a simple agent built upon the publicly available GPT-5.5 Pro model. The agent autonomously and robustly disproves the Erdős unit distance conjecture, successfully generating a correct proof in 8 out of 8 independent trials. Furthermore, it operates efficiently, consuming an average of approximately 107k reasoning tokens per trial. This demonstrates that, in hindsight, achieving such a breakthrough falls well within the capabilities of models publicly released prior to OpenAI's announcement, without relying on non-public models or complex scaffolding. To facilitate both the development of AI systems for mathematical reasoning and the use of AI in mathematical discovery, we release the agent's source code, all intermediate reasoning logs, and the 8 generated proofs on GitHub at `https://github.com/yichenhuang/unit-distance`.

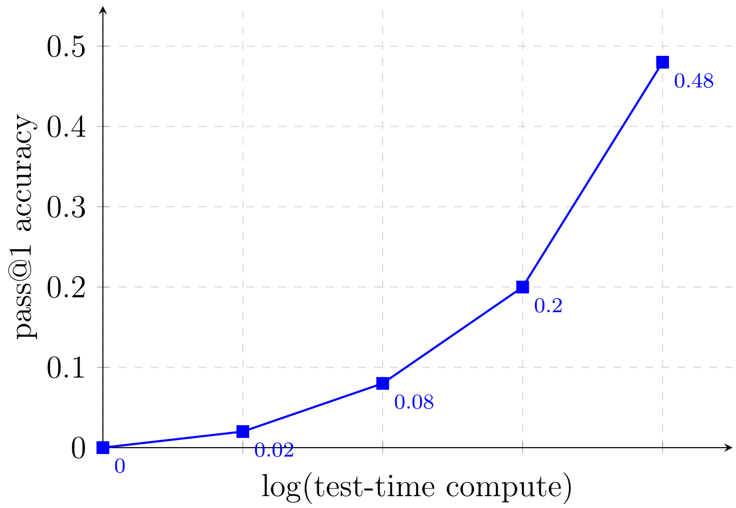

Figure 1: Test-time scaling curve for OpenAI's internal model, reproduced from [1]. The horizontal axis represents the logarithm of test-time compute (with the exact scale undisclosed), and the vertical axis represents the success probability. As shown, the pass@1 accuracy reaches 0.48 at the maximum test-time compute reported by OpenAI.

2 Agent

While termed an agent, our system employs a minimalist architecture: it runs a three-round conversation, prompting the underlying model to sequentially brainstorm a strategy, execute it, and critically review the resulting proof.

The system prompt establishes the context and the objective:

System Prompt

```
The provided theorem was recently proved, and the result is now accepted by the
mathematical community. Your task is to independently construct a proof.
```

The first prompt introduces the theorem statement and initiates the brainstorming phase:

First Prompt

```
**Theorem.** There exist an absolute constant $\delta > 0$ and an infinite
sequence $\{P_i\}$ of finite point sets in $\mathbb{R}^2$ such that $|P_i| \to
\infty$ and $u(P_i) \ge |P_i|^{1+\delta}$ for all $i$, where $u(P_i) =
\frac{1}{2} | \{ (p,q) \in P_i \times P_i : \|p-q\|_2 = 1 \} |$ is the number
of unit-distance pairs.

**Remark.** This result disproves the Erdős Unit Distance Conjecture.
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As a first step, brainstorm and propose three most promising approaches to
prove this theorem. Explore each approach to sufficient depth, and then rank
them from most to least promising.
```

The second prompt directs the model to execute its chosen strategy:

Second Prompt

```
Thoroughly develop your top-ranked approach to construct a complete and
rigorous proof, adapting your strategy as needed.
```

The third prompt initiates a critical self-review and finalizes the output:

Third Prompt

```
Critically examine the proof you constructed to identify and resolve any errors
or logical gaps. Then, present a complete, rigorous, step-by-step proof.
Clearly justify each step with sufficient detail so that an expert can verify
your reasoning without needing to fill in any gaps.
```

Two remarks are in order. First, the initial prompt directly asks the model to disprove the Erdős unit distance conjecture. Although this provides the correct direction, it supplies only a single bit of information. If it were unknown whether the conjecture is true or false, this hint could be removed simply by instructing the agent to attempt both a proof and a disproof in separate runs, effectively doubling the test-time compute. Second, the problem-specific text is confined entirely to the first half of the initial prompt. The reasoning instructions—brainstorming, executing, and reviewing—are problem-agnostic. Consequently, this pipeline can be applied to other theorem-proving tasks, requiring only minor modifications for problems that seek a final answer supported by a rigorous argument.

All trials were conducted via the GPT-5.5 Pro API. Tools and web search were disabled. The reasoning effort parameter was set to “xhigh”. The verbosity parameter was set to “high” for the first two rounds, and left at the default (medium) setting for the third round. All other parameters, including temperature and maximum output tokens, were kept at their default values. Crucially, since web search was disabled and the GPT-5.5 Pro model was released prior to OpenAI’s announcement, we can be certain there is no data contamination: the model’s training data cannot contain the recent proofs.

3 Results and Analysis

3.1 Performance and Efficiency

Our agent demonstrated consistent performance, successfully generating a correct proof in all 8 independent trials (a pass rate of 100% for the 8 trials). On average, the agent consumed 106.7k reasoning tokens per trial. The computational cost across the three stages of the pipeline is detailed in Table 1.

Table 1: Average reasoning token consumption and standard deviation across the 8 trials, broken down by pipeline stage. The brainstorming and execution phases required similar amounts of computation, whereas the final review phase used the fewest tokens. All token counts are rounded to the nearest 0.1k.

Phase	**Mean (k)**	**Standard Deviation (k)**
Round 1: Brainstorming	42.4	16.0
Round 2: Execution	41.9	11.1
Round 3: Review	22.3	3.8
Total	**106.7**	**22.7**

Directly comparing the efficiency of our agent with OpenAI's internal model is impossible due to the lack of publicly disclosed information regarding the amount of test-time compute used to generate the scaling curve in Figure 1. However, we can establish a lower-bound estimate using the abridged chain-of-thought document [11] linked in OpenAI's announcement. By extracting the text from this PDF and processing it through OpenAI's official token counting API (`client.responses.input_tokens.count`), we calculate the length of the published reasoning trace to be approximately 113k tokens.

This estimation carries two important caveats. First, text extraction from a mathematically dense PDF is imperfect, introducing a minor error in the calculated token count (estimated relative error $\lesssim 5\%$). Up to this uncertainty, this 113k token count serves as a lower bound because the PDF is explicitly described by OpenAI as an abridged version of the model's chain of thought. Second, this single published sample was likely curated and may not be representative of the average token consumption across all runs.

Nevertheless, this information provides useful context. OpenAI's internal model achieved a maximum success probability of 0.48 at an undisclosed, presumably large compute budget. In contrast, our GPT-5.5 Pro agent successfully produced a correct proof in all 8 trials using a structured prompting pipeline, with an average total reasoning length of 106.7k tokens. Since this average consumption is less than the 113k-token lower bound estimated for OpenAI's abridged chain of thought, it indicates that our autonomous agent likely operates at a lower computational cost while reliably generating correct proofs.

3.2 Proof Analysis

The proofs generated by our agent, as well as those produced by the OpenAI internal model [6] (and its human-digested version [7]) and Anthropic's non-public Claude Mythos model

[12], share a high-level strategy of reducing the geometric problem to algebraic number theory. However, they exhibit diversity in their mathematical realizations. We can classify these proofs based on two primary features: the specific number-theoretic input used to generate a large number of unit distances, and the geometric structure of the resulting planar point set.

Regarding the number-theoretic input, the OpenAI proof relies on an infinite tower of CM (Complex Multiplication) number fields in which at least one rational prime is completely split. In contrast, the Claude Mythos proof does not use CM fields or require a completely split prime. Instead, it relies on an infinite tower of mixed-signature number fields, generating unit-distance directions from algebraic integer solutions to a circle equation.

Regarding the point set structure, OpenAI's counterexample is constructed by restricting a high-dimensional lattice to a bounded window and projecting it injectively to the two-dimensional plane. The resulting point set does not possess a Cartesian product structure. Conversely, Claude Mythos's counterexample is explicitly constructed as a Cartesian product $X \times X$, where X is a finite subset of $\mathbb{R}$.

We label the 8 proofs autonomously generated by our GPT-5.5 Pro agent as Proof 1 through Proof 8, corresponding directly to their file names in our GitHub repository. Based on the two features discussed above, we classify these proofs alongside the OpenAI and Claude Mythos proofs in Table 2. It is important to note that while the proofs are categorized by these broad features, all proofs listed in the table are mathematically distinct. For instance, although Proofs 2, 3, 4, and 5 share the same overarching structure as the OpenAI proof, they differ from it and from each other in their specific mathematical realizations. Furthermore, our agent independently discovered novel hybrid configurations. Proofs 1, 6, and 8 utilize CM fields but assemble the points using a Cartesian product, whereas Proof 7 utilizes mixed-signature fields but produces a non-Cartesian point set.

Table 2: AI-generated proofs disproving the Erdős unit distance conjecture, categorized by their underlying algebraic number theory input and the geometric structure of the resulting planar point sets.

Proof Source	**Number-Theoretic Input**	**Geometric Structure**
OpenAI [6, 7], Proofs 2, 3, 4, 5	CM field tower, split prime	Non-Cartesian
Claude Mythos [12]	Mixed-signature field tower	Cartesian product
Proofs 1, 6, 8	CM field tower, split prime	Cartesian product
Proof 7	Mixed-signature field tower	Non-Cartesian

Disclosure of AI Assistance

Beyond using GPT-5.5 Pro for the autonomous agent, GPT-5.5, GPT-5.5 Pro, and Gemini 3.1 Pro Preview assisted the author in understanding dense mathematical literature and the proofs generated by the agent. GPT-5.5 Codex generated the agent's code and estimated the token count of OpenAI's abridged chain-of-thought document [11]. Gemini 3.1 Pro Preview drafted this paper based on materials provided by the author and assisted with editing.

The human author independently verified all proofs and retains full responsibility for the accuracy and integrity of the final content.

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